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WZ-Plan: A Roadway Work-Zone Testbed for Autonomous Vehicle Motion Planning via Reinforcement Learning

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Equal contribution

About this file

This file reserves the project's Paper link while the manuscript PDF is being finalized. It is not a preprint or a citable version of the paper.

Project overview

Autonomous driving has made substantial progress under general driving conditions, but roadway work zones remain challenging for motion planning. Work zones temporarily change lane layouts and restrict available road space, complicating interactions with surrounding vehicles and pedestrians, while limited datasets dedicated to work-zone motion planning hinder planner training and evaluation. Simulation supports controlled training and evaluation in safety-critical scenarios, while reinforcement learning allows planners to learn from interactions with the environment. We introduce WZ-Plan, a configurable CARLA-SUMO testbed comprising a normal-traffic work zone and five safety-critical scenario families - curved work zones, constrained corridors, jaywalker encounters, dense traffic, and opposing-lane closures - and demonstrate its use with a jointly trained Proximal Policy Optimization (PPO) planner and structured scene encoder. Scenario-wise evaluation characterizes planner capabilities, failure modes, and within-town transfer limitations, illustrating WZ-Plan's value as a platform for systematic development and evaluation of work-zone motion planners.

Project website: <https://shuo9898-zs.github.io/wz-plan/>

Source code: <https://github.com/shuo9898-zs/wz-plan>